

Sam Stryker

Chappaqua, New York | samueltownshendstryker@gmail.com | linkedin.com/in/sam-stryker | samstryker.github.io

SUMMARY

Incoming University of Chicago student with advanced preparation in physics, mathematics, and computer science, pursuing original research in quantum error mitigation in collaboration with a research scientist at IBM T.J. Watson Research Center. Recognized with a 3rd place finish in Computer Science at Regeneron WESEF research competition and both regional and national academic awards. Experience includes hardware design and detector construction, alongside independent projects in RF systems and software. In parallel, service as a certified volunteer firefighter and EMT has strengthened teamwork, resilience, and leadership under pressure. Studying physics and molecular engineering with a focus on quantum engineering, with the goal of advancing quantum technology to the point of possible widespread adoption.

EDUCATION

The University of Chicago, Chicago, IL *Class of 2030 (matriculating September 2026)*
Intending B.S. Physics & Molecular Engineering (double major), Quantum Engineering Track

- Pathways in Molecular Engineering (MENG 10100), credit-bearing summer class: accelerated 3-week course condensing 9 weeks of undergraduate content: engineering design, molecular modeling, density functional theory, nanomaterials, wet-lab and HPC components - Grade 100/100

Sungkyunkwan University, Seoul, South Korea *June 2026 - July 2026*
International Summer Semester — Statistics in Python, Korean

- Selected to deliver student graduation address, youngest student in the program

Horace Greeley High School, Chappaqua, NY *September 2022 - June 2026*
Three-Year Science Research Program, GPA: 4.0 / 4.0

- AP Physics C (Calculus-based Mechanics and E&M), AP Chemistry, AP Calculus BC
- Advanced Topics in College Mathematics (Calculus III + Linear Algebra), Advanced Topics in College Computer Science (Advanced Java data structures and algorithms)
- Activities: Science Olympiad, FIRST Robotics, Varsity Track & Field, Cross Country, Peer Tutor
- Honors: Summa Cum Laude; NYS Attorney General's Triple "C" Award; AP Scholar with Distinction x2; National Top School Recognition Award

RESEARCH EXPERIENCE

Independent Researcher (affiliated with IBM T.J. Watson Research Center's Dr. Malcolm Carroll) *2023 - 2025*
Self-directed: designed the methodology and research question, wrote all code, and ran all experiments independently; consulted with mentor roughly every two months

Phase I: A Combination of Bayesian Inference with Zero-Noise Extrapolation for Quantum Error Mitigation

- Built a Bayesian correction framework using classically simulated, calibration-informed priors (from *ibm_kyiv* day-of-experiment calibration data) to regularize noisy hardware measurements before zero-noise extrapolation
- On a 3-qubit test circuit on IBM's *ibm_kyiv* backend, reduced mean absolute error by 87% relative to standard ZNE (0.040 vs. 0.306), improved extrapolation fit from $R^2 = 0.962$ to 0.9999, and reduced measurement variability by 91–95%
- Completed technical research paper - Code available at github.com/samstryker/Bayesian-ZNE-Phase-1
- Recognition: 3rd Place - Computer Science | Regeneron WESEF (760+ competitors, largest regional ISEF-affiliated fair in the U.S.), 1st Place - Computer Science | Somers Science Research Fair (700+ competitors)

Phase II: Shadow-Informed Priors for Bayesian Zero-Noise Extrapolation

- Extended the framework with priors derived from quantum shadow tomography rather than classical simulation, removing the ~50-qubit classical-simulation ceiling of Phase I
- Benchmarked on H_2 and LiH VQE circuits on IBM's 133 qubit *ibm_torino* backend using Hamiltonian Monte Carlo posterior inference; shadow-derived priors failed to outperform standard ZNE in raw accuracy under weak noise-scaling, but the framework delivered full Bayesian uncertainty quantification, hardware-specific noise characterization, and quantitative diagnostics of prior-likelihood mismatch
- Technical paper available upon request

CERN Beamline for Schools Project Lead *2026*

- Organized and led a 5-student team where I authored and submitted a proposal to study radiation-induced spin-coherence degradation in NV-center CVD diamond via GeV hadron irradiation on CERN's T9 beamline
- Secured a letter of support and pre/post-irradiation characterization commitment from Prof. Carlos Meriles (CCNY); The Meriles Group has since expressed interest in a possible future summer research opportunity, contingent on the lab's funding

WORK EXPERIENCE

Firefighter - Chappaqua Volunteer Fire Department, Chappaqua, NY *January 2024 - Present*

- Certifications: Firefighter I | Hazardous Materials First Responder, Exterior/Interior Firefighting Operations, Firefighter Survival/Self-Rescue, Certified Emergency Vehicle Operator, FEMA ICS/NIMS training (IS-100/200/700)
- Communicate clearly and make time-critical decisions with incomplete information while coordinating team-based operations during live fire and rescue emergencies, applying standardized incident-command and communication protocols
- Train recruits in firefighting techniques and tactics preparing students of all ages for Firefighter I certification

Emergency Medical Technician (EMT) - Chappaqua EMS + Pleasantville EMS

December 2025 - Present

- New York State-certified EMT (NYS DOH, AHA BLS Provider)
- Lead patient care and direct and EMT aider while coordinating scene activities and communicating with dispatch, hospital staff, and other first-response personnel

Mathematics Instructor - Mathnasium

December 2024 - January 2026

- Teaching K-12 students mathematics from elementary concepts through calculus
- Develop adaptive instructional methods for individual learning needs and cognitive differences
- Improved ability to communicate technical concepts to diverse audiences
- Entrusted with the supervision and care of children of all ages

RELEVANT TECHNICAL PROJECTS

Scintillation-Based Cosmic-Ray Muon Detector

June, 2025

- Fabricated high-energy particle detection system using BC-408 scintillator and silicon photomultiplier readout, custom-designed and surface-mount soldered the signal-processing and peak-detection PCB

RF Security & Wireless Protocol Analysis

2023 - Present

- Designed and constructed rolljam device to analyze automotive rolling-code cryptography vulnerabilities and wireless security weaknesses
- Conducted comprehensive protocol analysis using software-defined radio (SDR) platforms and WiFi analysis tools
- Investigated security vulnerabilities in WiFi, NFC, radio frequency, and RFID communication protocols

ADDITIONAL HONORS & CERTIFICATIONS

- NSLI-Y Scholarship Recipient (Korean) | U.S. Department of State - one of 83 national finalists selected from 4,000+ application to receive the full scholarship
- IBM Quantum Learning: Basics of Quantum Information, Fundamentals of Quantum Algorithms
- IBM Qiskit Global Summer School 2026 - Advanced Track

SKILLS

Computational - Python (Qiskit, NumPy, SciPy, Pandas, Matplotlib), Java (data structures, algorithms, object-oriented programming), Linux/Bash (Ubuntu/Debian), Git/GitHub (version control), Fusion 360, AutoCAD

Quantum Computing - IBM Qiskit, Variational quantum algorithms (VQE), Quantum error mitigation (ZNE, Bayesian methods)